AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

T216B
Cisco Networking (CCCNA) part B

Midterm Examination (MTA)
Second Semester 2025/2026

Date: 16/Apr./2026

Number of Exam Pages: 5 (including this cover sheet)	**Time Allowed:** 2h

Instructions:

- Total Marks: **50**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and** ID on the External Answer booklet.
- **Calculators** are allowed.

جامعة العربية المفتوحة - مصر
Arab Open University
2025/2026 Spring
الفصل الدراسي
الامتحانات
Arab Republic of

CamScanner

Part 1: Multiple choices questions- 1 mark per question **Total: 5 Marks**

1. What device is considered a supplicant during the 802.1X authentication process?
   a. The authentication server that is performing client authentication
   b. The router that is serving as the default gateway
   c. The client that is requesting authentication
   d. The switch that is controlling network access
2. A network administrator of a college is configuring the WLAN user authentication process. Wireless users are required to enter username and password credentials that will be verified by a server. Which server would provide such service?
   a. RADIUS
   b. SNMP
   c. AAA
   d. NAT
3. What type of wireless antenna is best suited for providing coverage in large open space, such as hallways or large conference rooms?
   a. Dish
   b. Omnidirectional
   c. Yagi
   d. Directional
4. In FHRP terminology, what represents a set of routers that present the illusion of a single router to hosts?
   a. Virtual router
   b. Standby router
   c. Default gateway
   d. Forwarding router
5. What route would have the lowest administrative distance?
   a. A route through the OSPF routing protocol
   b. A static route
   c. A route received through the EIGRP routing protocol
   d. A route received through the RIP routing protocol

Part 2: Answer all the questions – 4 marks per question **Total: 16 Marks**

1. Answer the following questions:
   a. How does the First Hop Redundancy Protocol operate? (1 marks)
   b. What is the default HSRP priority? (1 mark)
   c. If two routers have the same HSRP priority; on which base can the two router elect the active router?
   d. If the HSRP election process is done; what will be the consequence if a new router with a higher priority and preemption enabled joins the network? (1 marks)
2. Answer the following questions:
   a. What is the role of the IEEE 802.1X standard? (1 mark)
   b. Which AAA component is responsible for determining what the user can access? (1 mark)
   c. How can we mitigate MAC address table attacks? (1 mark)
   d. How can we mitigate the exploitation of CDP? (1 mark)
3. Answer the following questions:
   a. How can we mitigate Spanning Tree protocol attacks? (1 mark)
   b. What should be done on to unused switch port in a company that adheres to the best security practices? (1 mark)
   c. An administrator who is troubleshooting connectivity issues on a switch, notices that a switch port configured for port security is in the err-disabled state. After verifying the cause of the violation, how should the administrator re-enable the port? (1 mark)
   d. How can VLAN attacks be mitigated? (1 mark)

Arab Open University
القبول والتسجيل
الامتحانات
Arab Republic of Egypt
CamScanner

4. Answer the following questions related to WLAN:
   a. What is the difference between autonomous APs and Controlled based APs? (2 marks)
   b. Explain how the CSMA/CA operates. (2 marks)

Part 3: Answer all the following questions **Total: 29 Marks**

1. Answer the following questions (10 marks)

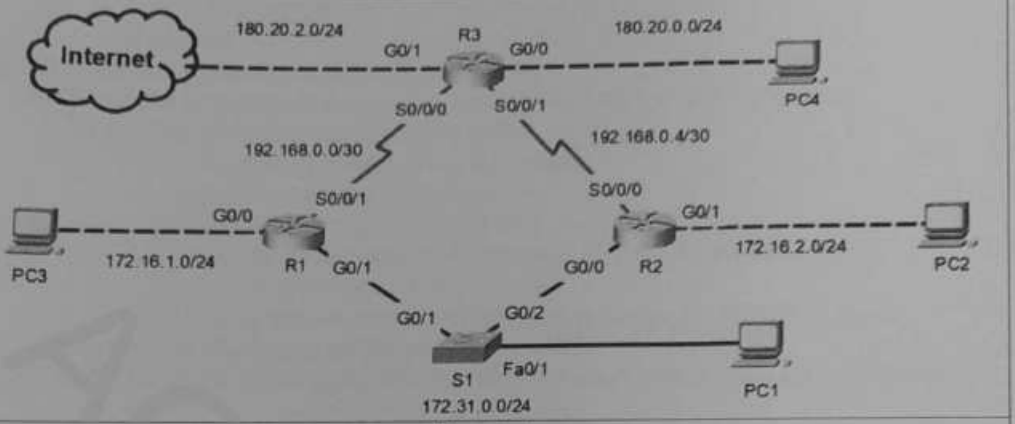


Figure 1

a) How many networks are directly connected to R3? (1 mark)
b) How many remote networks are there for R3? (1 mark)
c) On router R3 write the command to configure a: (4 marks)
   - static route to 172.16.1.0/24 using the exit interface S0/0/0
   - static route to 172.31.0.0/24 using the next hop IP address 192.168.0.6
d) On router R3 write the command to configure a floating static route to 172.31.0.0/24 using the exit interface S0/0/0 to be used as a back-up in case the one configured via R2 fails. (2 marks)
   Note: the administrative distance is 32
e) On router R3 write the command to configure a default static route to route all traffic to the Internet using the next hop IP address 180.20.2.0/24 (2 marks)

2. Refer to Figure 2 and answer the questions. (8 marks)

```
Gateway of last resort is not set

     15.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
C       15.0.140.0/30 is directly connected, Serial0/2/0
L       15.0.140.2/32 is directly connected, Serial0/2/0
C       15.0.140.4/30 is directly connected, Serial0/2/1
L       15.0.140.5/32 is directly connected, Serial0/2/1
O       15.0.140.8/30 [110/128] via 15.0.140.6, 00:03:10, Serial0/2/1
     172.16.0.0/26 is subnetted, 1 subnets
O       172.16.140.0/26 [110/65] via 15.0.140.1, 00:03:10, Serial0/2/0
     172.17.0.0/26 is subnetted, 1 subnets
O       172.17.140.0/26 [110/129] via 15.0.140.6, 00:03:10, Serial0/2/1
     174.10.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       174.10.140.0/25 is directly connected, GigabitEthernet0/0
L       174.10.140.1/32 is directly connected, GigabitEthernet0/0
```


Arab Open University
القبول والتسجيل
الامتحانات
Arab Republic of Egypt
CamScanner

Figure 2

a. How many directly connected networks does this route have? (1 marks)
b. List all the directly connected networks. (2 marks)
c. How many networks were learned dynamically? (1 marks)
d. List all the networks that were learned dynamically by R1. (2 marks)
e. Via which interface will the router forward the packet if the destination address is:
    a. 172.17.140.10
    b. 172.10.140.55
    (2 marks)

3. Answer the following questions: (11 marks)
   a) Write the commands needed to configure FHRP on R3. (5 marks)
   To do this complete the following tasks:
   - Access interface G0/1/2 (1 mark)
   - Set the version to 2 (1 mark)

   For the rest use group number 3
   - Set the router virtual IP address 177.50.30.9 (1 mark)
   - Set the priority to 180 (1 mark)
   - Force the re-election of the active router on HSRP (1 mark)

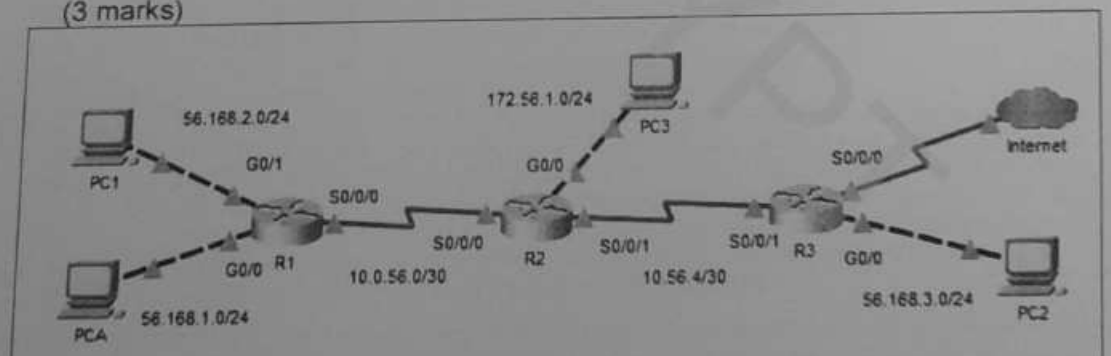
   The prompt displayed on the screen is: **R5#**

   b) Refer to figure 3 and complete the configuration of port security. (3 marks)

```
1 - interface FastEthernet0/16
2 - switchport mode _____
3 - switchport _____
4 - switchport port-security _____ 10
5 - switchport port-security _____ AC09.1234.17E3
6 - switchport port-security _____ sticky
7 - switchport port-security _____ shutdown
```

Figure 3

   c) Refer to figure 4 and explain why PC1, PCA, and PC3 are not able to connect to the internet while only PC2 is able to. Furthermore, it is noted that all the four PCs (PC1, PCA, PC3 and PC2) are able to successfully ping each other. (3 marks)



Router 1

الأكاديمية العربية المفتوحة
Arab Open University
القبول والتسجيل
الامتحانات
Arab Republic of Egypt

CamScanner

```
R1#do show ip route
        <output omitted>

      10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C        10.0.56.0/30 is directly connected, Serial0/0/0
L        10.0.56.2/32 is directly connected, Serial0/0/0
O        10.0.56.4/30 [110/128] via 10.0.56.1, 00:30:30, Serial0/0/0
      56.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
C        56.168.1.0/24 is directly connected, GigabitEthernet0/0
L        56.168.1.1/32 is directly connected, GigabitEthernet0/0
C        56.168.2.0/24 is directly connected, GigabitEthernet0/1
L        56.168.2.1/32 is directly connected, GigabitEthernet0/1
O        56.168.3.0/24 [110/129] via 10.0.56.1, 00:30:30, Serial0/0/0
      172.56.0.0/24 is subnetted, 1 subnets
O        172.56.1.0/24 [110/65] via 10.0.56.1, 00:30:30, Serial0/0/0
S*    0.0.0.0/0 is directly connected, Serial0/0/0
```

Router 2

```
R2#show ip route
        <output omitted>

      10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C        10.0.56.0/30 is directly connected, Serial0/0/0
L        10.0.56.1/32 is directly connected, Serial0/0/0
C        10.0.56.4/30 is directly connected, Serial0/0/1
L        10.0.56.5/32 is directly connected, Serial0/0/1
      56.0.0.0/24 is subnetted, 3 subnets
O        56.168.1.0/24 [110/65] via 10.0.56.2, 00:08:16, Serial0/0/0
O        56.168.2.0/24 [110/65] via 10.0.56.2, 00:15:30, Serial0/0/0
O        56.168.3.0/24 [110/65] via 10.0.56.6, 00:15:30, Serial0/0/1
      172.56.0.0/16 is variably subnetted, 2 subnets, 2 masks
C        172.56.1.0/24 is directly connected, GigabitEthernet0/0
L        172.56.1.1/32 is directly connected, GigabitEthernet0/0
S*    0.0.0.0/0 is directly connected, Serial0/0/0
```

Router 3

```
R3#show ip route
        <output omitted>

      10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C        10.0.56.0/30 is directly connected, Serial0/0/0
L        10.0.56.1/32 is directly connected, Serial0/0/0
C        10.0.56.4/30 is directly connected, Serial0/0/1
L        10.0.56.5/32 is directly connected, Serial0/0/1
      56.0.0.0/24 is subnetted, 3 subnets
O        56.168.1.0/24 [110/65] via 10.0.56.2, 00:08:16, Serial0/0/0
O        56.168.2.0/24 [110/65] via 10.0.56.2, 00:15:30, Serial0/0/0
O        56.168.3.0/24 [110/65] via 10.0.56.6, 00:15:30, Serial0/0/1
      172.56.0.0/16 is variably subnetted, 2 subnets, 2 masks
C        172.56.1.0/24 is directly connected, GigabitEthernet0/0
L        172.56.1.1/32 is directly connected, GigabitEthernet0/0
S*    0.0.0.0/0 is directly connected, Serial0/0/0
```

Figure 4

End of questions

الجامعة العربية المفتوحة
Arab Open University
Arab Republic of Egypt

CamScanner